



**Department of Electrical and Computer Engineering
North South University**

Senior Design Project

Automated Bias Mitigation & Multi-Source News Aggregation

Submitted by:

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LETTER OF TRANSMITTAL

December, 2025

To

Prof. Mohammad Abdul Matin

Chairman

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: Submission of Capstone Project Report on

“Automated Bias Mitigation & Multi-Source News Aggregation”

Dear Sir,

With due respect, we would like to submit our Capstone Project Report entitled “Automated Bias Mitigation & Multi-Source News Aggregation” as a partial fulfillment of the requirements of CSE 499A of our Bachelor of Science in Engineering program.

This project focuses on developing an automated system that collects news from multiple sources, detects subjective bias using artificial intelligence, and generates a neutralized version of news content while preserving factual accuracy. The project has been a valuable learning experience for us, as it allowed us to apply theoretical knowledge of machine learning, natural language processing, and ethical AI to a real-world problem.

We have tried our best to complete this project with maximum sincerity and competence, ensuring that all academic and technical requirements have been met. We sincerely hope that this report will meet your expectations and serve its intended academic purpose.

We shall be highly obliged if you kindly accept this report and provide your valuable judgment.

Sincerely yours,

Tashfia Adiba Zaman Nizhum

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APPROVAL

Nehla Nujhath Neha (ID: 2122234642) and Tashfia Adiba Zaman Nizhum (ID: 2131228042) from the Department of Electrical and Computer Engineering, North South University, have worked on the Senior Design Project titled “Automated Bias Mitigation & Multi-Source News Aggregation” under the supervision of Dr. Mohammad Ashrafuzzaman Khan (AzK) as a partial fulfillment of the requirements for EEE/CSE 499A and the degree of Bachelor of Science in Engineering. This project has been accepted as satisfactory.

Supervisor’s Signature

.....

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Chairman’s Signature

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Prof. Mohammad Abdul Matin

Chairman

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DECLARATION

This is to declare that the project titled “Automated Bias Mitigation & Multi-Source News Aggregation” is our original work. No part of this work has been submitted, either partially or fully, for the award of any other degree or diploma at any institution.

All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works referred to in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been strictly maintained throughout the preparation of this project.

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ABSTRACT

Automated Bias Mitigation & Multi-Source News Aggregation

The rapid growth of digital media has increased public access to news, but it has also intensified the presence of subjective framing, sensational language, and partisan bias, which can mislead readers and influence public opinion. The current news consumption landscape lacks effective automated mechanisms to deliver neutral and fact focused information from diverse sources. This project presents an automated system designed to collect, analyze, and neutralize news articles obtained from multiple reliable sources in order to provide objective and balanced news content.

The proposed system follows a structured pipeline that retrieves news articles, extracts relevant textual content, evaluates bias at the sentence level, and reduces emotionally charged language while preserving factual information. The methodology emphasizes extracting verifiable events rather than opinion-based commentary. The system applies linguistic and contextual analysis techniques to identify biased expressions and iteratively refines the content to meet predefined neutrality standards.

Experimental results demonstrate that the system successfully removes subjective and inflammatory language while maintaining the core meaning of the news events. The project contributes to improved media literacy, supports informed decision making, and promotes responsible news consumption by providing clear, neutral, and fact oriented information suitable for social, political, economic, and public interest applications

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Chapter 1 Introduction

1.1 Background and Motivation

In the digital age, news media plays a crucial role in shaping public opinion, influencing political decisions, economic behavior, and social attitudes. With the rapid expansion of online news platforms and social media, information is produced and disseminated at an unprecedented rate. However, modern news reporting is often affected by subjective framing, sensational headlines, emotional language, and ideological bias. Such bias can distort facts, amplify polarization, and reduce public trust in media outlets.

Contemporary journalism frequently blends factual reporting with opinionated commentary, making it difficult for readers to distinguish between objective events and subjective interpretations. This problem becomes more severe during politically sensitive periods, economic instability, public health emergencies, or national crises, where emotionally charged reporting can cause panic, misinformation, or misguided decision making. Readers are often exposed to news from a single source or ideological perspective, reinforcing confirmation bias rather than promoting balanced understanding.

Although multiple news aggregators exist, most simply collect and display articles without analyzing or mitigating bias. They fail to address emotionally loaded language or source redundancy, allowing biased narratives to persist. Therefore, there is a growing need for an automated system that not only aggregates news from multiple credible sources but also analyzes and neutralizes biased language while preserving factual integrity.

This project is motivated by the necessity to provide readers with clear, fact focused, and neutral news content. By reducing subjective framing and emphasizing verifiable events, the proposed system aims to support informed public discourse, enhance media literacy, and promote responsible consumption of news.

1.2 Purpose and Goal of the Project

The primary purpose of this project is to design and implement an automated pipeline for bias mitigation and multi-source news aggregation. The system aims to retrieve news articles from diverse and credible sources, analyze textual bias, and generate neutralized versions of news content that focus strictly on factual events.

The key goals of the project are as follows:

- To collect news articles from multiple trusted sources while ensuring source diversity and credibility.
- To identify subjective, emotionally charged, and opinion based language within news articles.
- To mitigate detected bias by systematically removing or refining biased modifiers while preserving essential factual information.
- To provide transparent and explainable outputs that allow users to understand how bias was identified and reduced.
- To promote ethical, fair, and responsible use of artificial intelligence in media analysis.

The novelty of this project lies in its integrated approach, combining multi-source aggregation, sentence-level bias quantification, explainable artificial intelligence, and iterative text neutralization within a single automated framework. Unlike traditional news aggregators, this system actively transforms content to reduce bias rather than merely presenting multiple viewpoints.

1.3 Organization of the Report

The remainder of this report is organized into several chapters to clearly present the development, implementation, and evaluation of the proposed project. Chapter 2 presents a review of related research and existing works in the fields of media bias, news aggregation, and automated text analysis. It also discusses the limitations of current approaches that motivate the need for the proposed system. Chapter 3 describes the methodology of the project, including the overall system design, hardware and software components, and implementation details of the proposed framework. Chapter 4 covers the investigation and experimentation carried out in this project. It presents the obtained results along with detailed analysis and discussion to evaluate the performance and effectiveness of the system. Chapter 5 discusses the impacts of the project, focusing on societal, health, safety, legal, and cultural aspects, as well as environmental and sustainability considerations. Chapter 6 outlines the project planning and budget, providing an overview of time management, resource allocation, and cost considerations. Chapter 7 addresses complex engineering problems and activities associated with the project, highlighting how the proposed system meets professional engineering requirements. Finally, Chapter 8 concludes the report with a summary of the project, identifies its limitations, and suggests possible directions for future improvements. The report ends with a list of references cited throughout the document.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

In recent years, several studies have explored media bias detection, sentiment analysis, and automated news aggregation using natural language processing and machine learning techniques. Researchers have proposed various approaches to identify biased language in news articles by analyzing sentiment polarity, lexical choices, and contextual features. Many of these works focus on classifying news articles as biased or unbiased using supervised learning models trained on labeled datasets.

For instance, Hamborg et al. proposed a news bias detection framework that analyzes political bias by examining word usage patterns and framing techniques in news articles. Their study demonstrated that linguistic features could effectively identify ideological bias; however, the approach mainly focused on classification rather than bias reduction. Similarly, Baly et al. developed a neural network based system to detect media bias using contextual embeddings, achieving improved accuracy compared to traditional machine learning methods. While effective in bias identification, the system did not provide mechanisms for content neutralization or explanation.

Other studies have addressed news aggregation by collecting articles from multiple sources to provide broader coverage. Most existing news aggregators emphasize source diversity and topic clustering but do not analyze the emotional or subjective nature of the content. Sentiment analysis-based approaches, such as those using transformer based language models, have been applied to identify emotionally charged language in news. However, these methods often operate as black box models, offering limited transparency and interpretability. After a detailed examination of the literature, the following limitations have been identified: (i) Most existing works focus primarily on bias detection or classification rather than automated bias mitigation or content neutralization. (ii) Many studies rely on single source or limited datasets, which restricts source diversity and may introduce dataset specific bias. (iii) Existing news aggregation systems generally lack explainable mechanisms to justify why certain content is considered biased. (iv) Few approaches attempt to preserve factual integrity while removing subjective or emotionally loaded language. These limitations motivated the development of the proposed project, which integrates multi-source news aggregation with automated bias mitigation and explainable analysis. Unlike existing works, the proposed system not only detects biased expressions but also refines the content iteratively to produce neutral, fact focused news while maintaining transparency and ethical responsibility.

Chapter 3 Methodology

3.1 System Design

The proposed system is designed as an automated and modular pipeline that performs multi-source news aggregation followed by bias detection, mitigation, and neutral content generation. The overall system architecture consists of sequential processing stages that ensure data credibility, neutrality, and transparency. Initially, the system performs multi-source data ingestion, where news articles are retrieved in real time using geo-targeted queries focused on the Bangladeshi news ecosystem. A source diversity enforcement mechanism is applied to ensure that articles originate from multiple trusted and distinct domains, preventing redundancy and source dominance. Next, the content extraction and parsing module processes the retrieved web pages. This module isolates the main article text by removing advertisements, navigation elements, and irrelevant metadata. The extracted text is then segmented into sentences for further analysis. The bias analysis module evaluates each sentence using a deep learning based contextual language model. A bias probability score is assigned to quantify the level of subjective or emotionally charged language. To ensure transparency, an explainable analysis component highlights the specific words responsible for high bias scores. Finally, the bias mitigation and refinement module removes or normalizes biased modifiers while preserving essential factual elements such as subjects, actions, and objects. The system applies iterative validation to ensure the refined content meets predefined neutrality thresholds. The final output is a neutral, fact focused news summary along with links to original sources. A block diagram illustrating the system workflow is included to visually represent the interaction between these modules.

3.2 Hardware and/or Software Components

This project is implemented entirely as a software based system and does not require any dedicated hardware components. The key software tools, libraries, and frameworks used in the development of the system are described below.

Software Components

Tool / Framework	Function	Other Similar Tools	Reason for Selection
SerpAPI	Real-time, geo-targeted news retrieval	NewsAPI, GDELT	Provides reliable location-based search results and source diversity
Newspaper3k	Article extraction and parsing	BeautifulSoup, Scrapy	Efficient extraction of clean article text

Tool / Framework	Function	Other Similar Tools	Reason for Selection
PyTorch	Deep learning framework	TensorFlow, Keras	Flexible model implementation and strong research support
Hugging Face Transformers	Pre-trained language model integration	AllenNLP	Easy access to state-of-the-art transformer models
DistilBERT	Contextual language understanding	BERT, RoBERTa	Lightweight and efficient with strong performance
Captum	Model interpretability	LIME, SHAP	Enables explainable bias analysis using integrated gradients
TextBlob	Linguistic and grammatical analysis	SpaCy	Simple and effective syntactic processing
NLTK	Tokenization and text processing	SpaCy	Reliable natural language preprocessing utilities

3.3 Hardware and/or Software Implementation

The software implementation follows a modular architecture to ensure scalability, maintainability, and transparency. The data acquisition module sends queries to the news retrieval interface and collects article URLs based on predefined filtering rules. The parsing module processes each article using text extraction libraries to obtain clean and structured content.

The bias detection module applies a pre-trained contextual language model to analyze sentence-level semantics and compute bias probability scores. Explainable analysis is implemented to identify high-impact tokens contributing to bias decisions. These insights are logged to ensure algorithmic transparency.

The bias mitigation module performs syntactic filtering by selectively removing or normalizing emotionally charged adjectives and adverbs while retaining factual nouns and verbs. The system employs an iterative refinement loop to validate neutrality until acceptable thresholds are achieved. All modules are integrated into a unified pipeline that outputs neutralized news content along with original source references. The system is designed to operate without storing user data, ensuring privacy and ethical compliance.

Chapter 4 Investigation, Experiment, Result, Analysis and Discussion

4.1 Experimental Setup

To evaluate the effectiveness of the proposed automated bias mitigation and multi-source news aggregation system, a series of controlled experiments were conducted using real-time news articles collected from multiple reputable online news sources.

The experiments focused on assessing bias detection accuracy, effectiveness of bias mitigation, and preservation of factual information. News articles were retrieved using geo-targeted queries to ensure relevance to the Bangladeshi news ecosystem.

Articles from different domains covering political, economic, social, and public interest topics were selected. Each article was processed through the complete system pipeline, including extraction, bias scoring, explainable analysis, and neutralization. Key variables examined during the experiments included bias probability score before and after mitigation, number of emotionally charged words removed, and sentence structure consistency after refinement.

4.2 Results

The experimental results demonstrate that the system effectively identifies and mitigates biased language while maintaining the core factual content of news articles.

Picture 1: We have tasted on a single line, which goes through an iteration loop. Then it gives us a biased report and removes terms and neutralizes lies.

```
Loading models... please wait.

Enter your news paragraph: The corrupt politician shamelessly lied about the massive scandal

--- Processing: 'The corrupt politician shamelessly lied about the massive scandal' ---
-> Iteration 0 Suspects: ['massive(0.26)', 'shamelessly(0.24)', 'scandal(0.21)', 'politician(0.08)']
  >>> DETECTED (Priority High): 'massive'
-> Iteration 1 Suspects: ['shamelessly(0.31)', 'scandal(0.26)', 'politician(0.11)', 'corrupt(0.09)']
  >>> DETECTED (Priority High): 'shamelessly'
-> Iteration 2 Suspects: ['scandal(0.55)', 'politician(0.31)', 'corrupt(0.21)', 'lied(0.15)']
  >>> DETECTED (Priority High): 'corrupt'
-> Iteration 3 Suspects: ['scandal(0.59)', 'politician(0.31)', 'lied(0.26)']
  >>> SKIP 'scandal': Appears to be a Head Noun/Object. Deletion unsafe.

[ Status ]
BIASED (Negative, Confidence: 99.58%)
Removed Terms: massive, shamelessly, corrupt

[ Neutralized Paragraph ]
The politician lied about the scandal.
```

Picture 2 and 3:

In the second screenshot, after we give a topic, it fetches the top 5 articles. And in the third screenshot, the article is tested, and we get the biased report and neutralize paragraph.

```
Enter news topic: metro rail expansion
🔍 Searching Latest News for: 'metro rail expansion'...
  found: MRT Line-1 slashed 91%, Line-5 down 60% ... (2 hours ago)
  found: Bangladesh's external debt jumps 42% in ... (1 day ago)
  found: 50 companies get permission to import 1,... (1 day ago)
  found: Four exporters honoured at HSBC Export E... (3 days ago)
  found: Metro commuters get digital top-up optio... (2 weeks ago)

✅ Found 5 recent articles. Extracting full text...

📄 Processing [1/5]: The Business Standard...

=====
SOURCE: The Business Standard
TITLE: MRT Line-1 slashed 91%, Line-5 down 60% in revised ADP
LINK: https://www.tbsnews.net/economy/mrt-line-1-slashed-91-line-5-down-60-revised-adp-1306926
-----
Highlights:

Government slashes major metro rail budgets due to stalled tenders

MRT-1 allocation cut over 90% amid cost review

MRT-5 Northern, airport, port projects face steep RADP reductions

Cost reassessment freezes tenders; Bangladesh metro costs unusually high

Roopdur unchanged: Ashulia expressway gets more funds for progress

[ Status ]
BIASED (Negative, Confidence: 99.59%)
Removed Terms: many, organisational, individual, other, main, last, widely, caretaker

[ Neutralized Paragraph ]
Following the fall of the Awami League government in the July uprising, of the party's top leaders have gone into hiding or taken refuge abroad.
```

Chapter 5: Impacts of the Project

5.1 Impact of the Project on Societal, Health, Safety, Legal, and Cultural Issues

The *Automated Bias Mitigation and Multi-Source News Aggregation* project has significant societal implications, particularly in promoting access to objective, factual, and balanced information. In an era where media bias, sensationalism, and misinformation are prevalent, this system helps reduce information asymmetry by extracting verifiable facts from multiple trusted news sources and neutralizing emotionally charged or opinionated language. As a result, the project enhances media literacy, reduces political and social polarization, and supports informed decision-making among citizens. From a health and safety perspective, unbiased news dissemination is crucial during emergencies such as pandemics, natural disasters, or public health crises. By removing panic-inducing language and focusing on factual reporting, the system enables the public to receive critical information calmly and accurately, reducing misinformation-driven fear, anxiety, and unsafe behavior. In terms of legal and ethical considerations, the project adheres to principles of algorithmic transparency and explainable artificial intelligence (XAI). By identifying and logging removed biased terms, the system allows users to understand how content is modified, ensuring accountability and trust. Furthermore, copyright and intellectual property rights are respected through transformative summarization and redirection to original news sources, aligning with fair-use practices. The stateless system architecture also ensures data privacy by avoiding the storage of personal user data or query histories. Culturally, the project supports inclusivity and fairness by minimizing ideological dominance and ensuring diverse perspectives are represented through enforced multi-source aggregation. This approach helps preserve cultural neutrality, encourages respectful discourse, and fosters a more informed and tolerant society.

5.2 Impact of the Project on Environment and Sustainability

As a **software-based project**, the environmental footprint of the Automated Bias Mitigation and Multi-Source News Aggregation system is relatively minimal compared to hardware-intensive solutions. However, its contribution to environmental sustainability is indirect yet meaningful. By promoting digital news consumption over physical newspapers, the system supports the reduction of paper usage, printing resources, and associated waste, contributing to environmental conservation. Additionally, by providing objective reporting on environmental issues such as climate change, pollution, and sustainability initiatives, the system empowers users with accurate information necessary for environmentally responsible decision-making. Neutralized and fact-based environmental news helps prevent misinformation and exaggeration, enabling policymakers, researchers, and the public to better understand real environmental conditions and challenges. The project also emphasizes computational efficiency through the use of optimized transformer models such as DistilBERT, which require fewer resources compared to larger models. This design choice supports sustainable computing practices by reducing energy consumption during large-scale text processing. Overall, the project demonstrates how intelligent software systems can contribute to sustainability by enabling responsible information dissemination, supporting digital efficiency, and fostering awareness of environmental issues through unbiased reporting.

Chapter 6: Project Planning and Budget

6.1 Project Planning (Gantt Chart)

The project was planned and executed over a period of approximately **11 weeks**, from **20 September 2025 to 06 December 2025**. The timeline was structured to ensure systematic progress from requirement analysis to final documentation and submission

Gantt Chart: Project Timeline (Sep 20 – Dec 06, 2025)

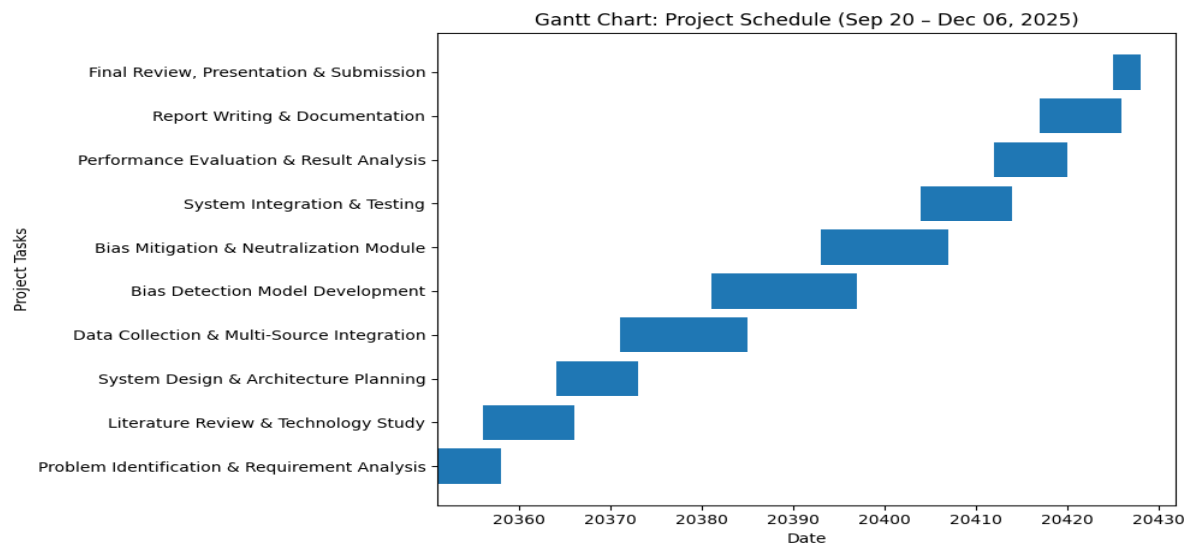


Figure 1: Gantt chart showing the project schedule from September 20 to December 06, 2025.

6.2 Project Budget

This project does not require a dedicated financial budget as it is entirely **software-based** and was developed using **open-source tools, freely available libraries, and existing hardware resources**. All development and testing activities were performed on personal computers already available to the project team. The software frameworks and tools used—such as Python, PyTorch, Hugging Face Transformers, Newspaper3k, NLTK, TextBlob, and Captum—are open-source and freely accessible. News data was retrieved using publicly available online sources with limited API usage within free-tier constraints. No proprietary software, specialized hardware, or paid cloud infrastructure was required for the successful completion of the project. As a result, the **overall project cost is considered negligible**, making the system cost-effective, scalable, and suitable for real-world deployment without additional financial investment.

Chapter 7: Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

The proposed project addresses several **Complex Engineering Problems (CEP)** as defined by modern engineering practice. The complexity arises from the need to process large-scale unstructured text data, mitigate bias using artificial intelligence, ensure transparency, and balance ethical, legal, and societal requirements. Table II summarizes the CEP attributes addressed by this project.

TABLE II. COMPLEX ENGINEERING PROBLEM (CEP) ATTRIBUTES

Attributes	Addressing the Complex Engineering Problems (P) in the Project
P1 Depth of knowledge required (K3–K8)	The project requires advanced knowledge of Engineering & IT tools (K6) , Natural Language Processing (NLP) , Machine Learning and Deep Learning models (DistilBERT) , Explainable AI (XAI) concepts, software system design , and analysis of scientific research papers (K8) related to bias detection and mitigation.
P2 Range of conflicting requirements	The project must balance news neutrality and factual accuracy , bias reduction vs. content readability , automation vs. transparency , and information completeness vs. legal and ethical constraints such as copyright and fair use.
P3 Depth of analysis required	There is no single optimal approach to bias detection and mitigation. Extensive analysis is required to choose appropriate models, token attribution methods, neutrality thresholds, and linguistic normalization strategies from multiple alternatives.
P4 Familiarity of issues	The project involves partially familiar and evolving problems , such as media bias, political framing, sentiment analysis, and misinformation, which vary across sources, regions, and cultural contexts.
P5 Extent of applicable codes and standards	There are no fixed engineering standards for automated bias mitigation in news systems. The project follows ethical AI guidelines , data privacy principles, and explainable AI best practices rather than rigid formal codes.
P6 Extent of stakeholder involvement	Multiple stakeholders are involved, including news consumers, journalists, researchers, policy makers, and platform developers , all of whom are affected by the system's neutrality and transparency.
P7 Interdependence	The project consists of interdependent subsystems such as multi-source data ingestion, text extraction, bias detection model, explainability

Attributes	Addressing the Complex Engineering Problems (P) in the Project
	module , and bias mitigation engine , where changes in one component directly affect others.

Table II demonstrates the Complex Engineering Problem attributes addressed by the project.

7.2 Complex Engineering Activities (CEA)

The project also involves **Complex Engineering Activities (CEA)** that require innovation, collaboration, modern engineering tools, and consideration of societal consequences. These activities are summarized in Table III.

TABLE III. COMPLEX ENGINEERING ACTIVITIES (CEA) ATTRIBUTES

Attributes	Addressing the Complex Engineering Activities (A) in the Project
A1 Range of resources	The project utilizes human resources (team collaboration) , modern software tools , open-source AI frameworks , NLP libraries , and online news data sources , without the need for specialized hardware.
A2 Level of interactions	The project requires continuous interaction among team members , consultation with supervisors , and consideration of end-user expectations , ethical implications, and societal impact.
A3 Innovation	The project introduces an innovative automated pipeline that not only detects bias but also explains and mitigates it , combining sentiment analysis , explainable AI , and linguistic normalization in a novel manner.
A4 Consequences to society / environment	The system has strong societal impact by reducing misinformation, polarization, and sensationalism. It supports responsible journalism , informed decision-making, and calm dissemination of critical information during emergencies.
A5 Familiarity	Engineers must be familiar with machine learning models , NLP techniques , data ethics , software development tools , and responsible AI practices. The project aligns with UN SDG #04 (Quality Education) by promoting media literacy and UN SDG #10 (Reduced Inequality) by ensuring fair access to unbiased information.

Chapter 8: Conclusions

8.1 Summary

This project presented the design and implementation of an **Automated Bias Mitigation and Multi-Source News Aggregation system** aimed at delivering factual, neutral, and objective news content. The system collects news articles from multiple credible sources, extracts relevant textual content, and applies natural language processing techniques to identify biased or emotionally charged language. A transformer-based model (DistilBERT) is used to compute bias probability scores at the sentence level, while explainable AI techniques are employed to identify the specific terms contributing to bias.

Based on the identified bias, the system performs linguistic normalization by removing or modifying subjective adjectives and adverbs while preserving core factual information. This iterative neutralization process ensures that the final output maintains informational integrity while minimizing emotional framing. The results demonstrate that the system successfully reduces sensationalism and partisan language, fulfilling the objective of delivering “just the facts” to users.

8.2 Limitations

Despite its effectiveness, the project has several limitations. First, bias detection is inherently complex and context-dependent; subtle ideological bias, sarcasm, or culturally embedded framing may not always be accurately identified by the model. Second, the system primarily focuses on **text-based news articles** and does not process multimedia content such as images, videos, or audio, which can also convey bias.

Additionally, the performance of the system depends on the quality and diversity of the input news sources. If source articles are incomplete or highly polarized, complete neutrality may be difficult to achieve. The use of pre-trained language models may also introduce **inherited biases** from training data, despite mitigation efforts. Finally, the system currently operates under limited real-time constraints and may face scalability challenges when processing very large volumes of news data simultaneously.

8.3 Future Improvement

Several enhancements can be considered to improve the system in the future. More advanced transformer models and **domain-adaptive fine-tuning** could be incorporated to better capture subtle and contextual forms of bias. Multilingual support can be added to extend the system beyond English-language news, enabling broader global applicability. Future versions may also include **multimodal bias detection**, integrating text with image and video analysis for a more comprehensive evaluation of media bias. User feedback mechanisms could be introduced to allow

readers to assess neutrality, enabling continuous system improvement through human-in-the-loop learning. Additionally, deploying the system as a scalable web or mobile platform with real-time processing capabilities would enhance accessibility and practical impact.

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